History of a Cell

Table 7.1: History of Cell

Year	Scientist	Events
1665	Robert Hooke	Coined word "Cell"
1670–74	Anthony van Leeuwenhoek	First living cells observed in microscope - Structure of bacteria
1831–33	Robert Brown	Presence of nucleus in cells of orchid roots
1839	Jan Evangelista Purkyne (J.E. Purkinje)	Coined "protoplasm"
1838–39	Schleiden & Schwann	Cell theory
1858	Rudolph Ludwig Carl Virchow	Cell theory '*omnis cellula e cellula*'
1873	Anton Schneider	Described chromosomes (Nuclear filaments) for the first time
1882	Walther Flemming	Coined the word mitosis; chromosome behaviour
1883	Edouard Van Beneden	Cell division in round worm
1888	Theodor Boveri	Centrosome; Chromosome Theory

7.2.1 Duration of Cell Cycle

Different kinds of cells have varied duration for cell cycle phases. Eukaryotic cell divides every 24 hours. The cell cycle is divided into mitosis and interphase. In a cell cycle 95% is spent for interphase whereas the mitosis and cytokinesis last only for an hour.

Table 7.2: Cell cycle of a proliferating human cell

Phase	Time duration (in hrs)
G_1	11
S	8
G_2	4
M	1

The different phases of cell cycle are as follows (Figure 7.1).

7.2.2 Interphase

Longest part of the cell cycle, but it is of extremely variable length. At first glance the nucleus appears to be resting but this is not the case at all. The chromosomes previously visible as thread like structure, have dispersed. Now they are actively involved in protein synthesis, at least for most of the interphase.

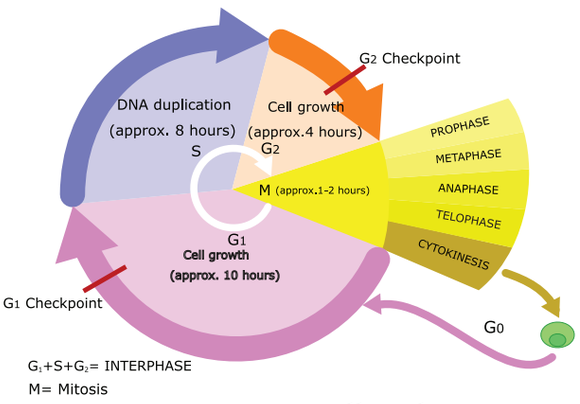


Figure 7.1: Cell cycle

C-Value is the amount in picograms of DNA contained within a haploid nucleus.

7.2.3 G_1 Phase

The first gap phase – 2C amount of DNA in cells of G_1. Cells become metabolically active and grows by producing proteins,

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